PROJECT REPORT[Designing Traffic Light Controller]

A Project Submitted

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(Section-A)

Under the Supervision of

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In Partial Fulfilment of

the requirements for the course of

CS1252 (Digital System Design Using HDL)

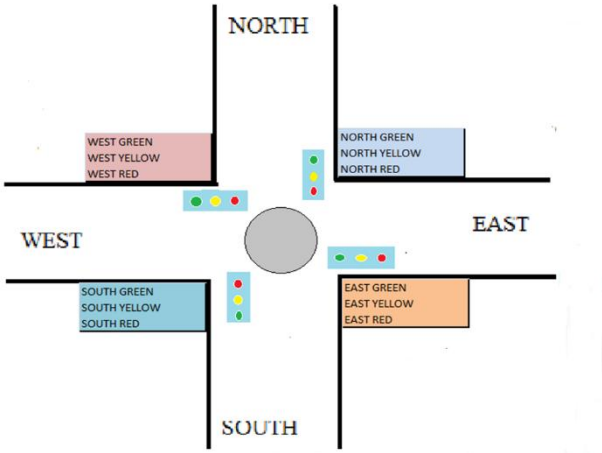
To the

Institute of Engineering And Technology (IET)

J.K Lakshmipat University Jaipur

INTRODUCTION:

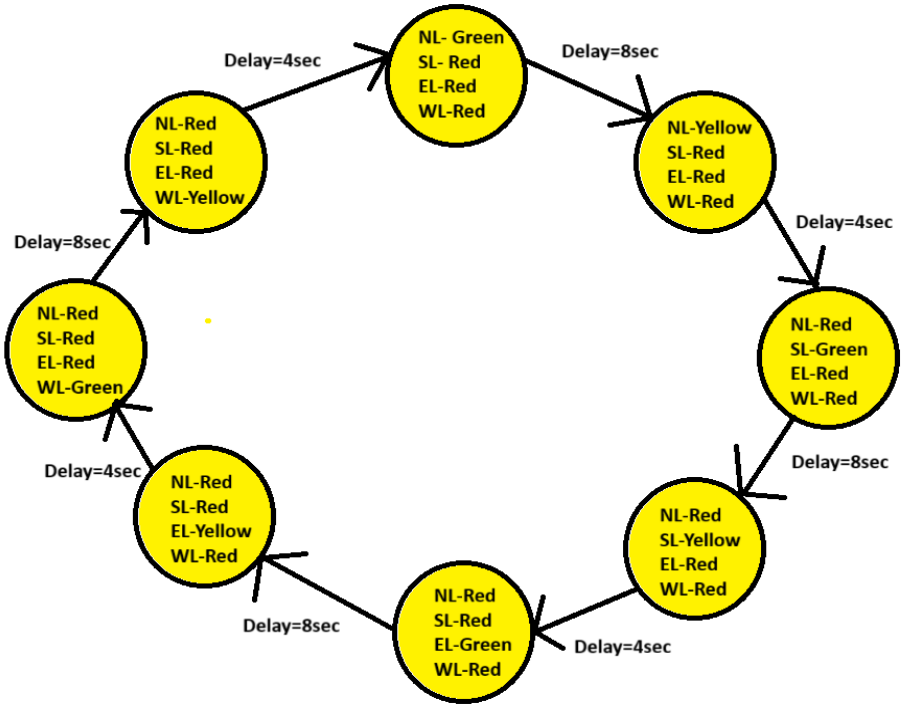
Traffic light controllers are an essential part of modern transportation systems, ensuring safe and orderly vehicle movement at road intersections. In digital system design, such controllers serve as excellent examples for understanding the application of finite state machines (FSMs), sequential logic, and timing-based control. This project focuses on designing and testing a **Traffic Light Controller** using digital design principles, where the system is driven by a clock signal and can be reset to an initial state.



The controller manages four directions North (NL), South (SL), East (EL), and West (WL) with each direction represented by a 3-bit output indicating the active light. According to the specification, "001" represents a Green light, "010" represents Yellow, and "100" represents Red. When the reset signal is asserted, the system transitions to the **North Enable (NE)** state, which acts as the starting state of the FSM. From there, the controller cycles through the sequence: North → South → East → West, repeating continuously.

Each direction receives a Green signal for eight clock cycles followed by a Yellow signal for four clock cycles before shifting control to the next direction. This project implements the entire logic using a structured state machine approach and verifies its functionality through simulation and testing. The design demonstrates the use of synchronous sequential logic to achieve predictable, safe, and time-controlled traffic management.

STATE DIAGRAM:



CHARACTERISTIC TABLE:

CURRENT STATE					NEXT STATE				
A	B	C	D	E	P/D1	Q/D2	R/D3	S/D4	T/D4
0	0	0	0	1	0	0	0	1	0
0	0	0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0	0	1
0	1	0	0	1	0	1	0	1	0
0	1	0	1	0	0	1	1	0	0
0	1	1	0	0	1	0	0	0	1
1	0	0	0	1	1	0	0	1	0
1	0	0	1	0	1	0	1	0	0
1	0	1	0	0	1	1	0	0	1
1	1	0	0	1	1	1	0	1	0
1	1	0	1	0	1	1	1	0	0
1	1	1	0	0	0	0	0	0	1

In the characteristic table of the Traffic Light Controller, the system signals are encoded using binary representations for simplicity and efficient digital implementation. The two-bit input AB is used to identify the active road direction among the four possible sides of the intersection. The encoding is as follows:

- AB = 00 → NL (North Light)
- AB = 01 → SL (South Light)
- AB = 10 → EL (East Light)
- AB = 11 → WL (West Light)

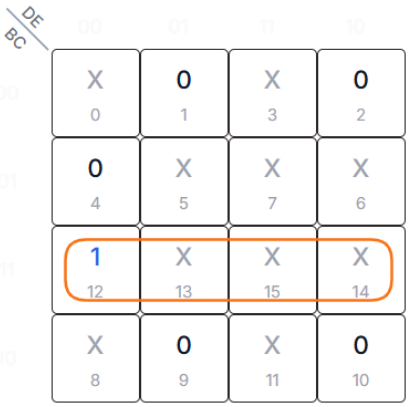
The three-bit signals CDE represent the status of the traffic light for the currently enabled direction. The output encoding follows the project specification:

- CDE = 001 → Green Light (G)
- CDE = 010 → Yellow Light (Y)
- CDE = 100 → Red Light (R)

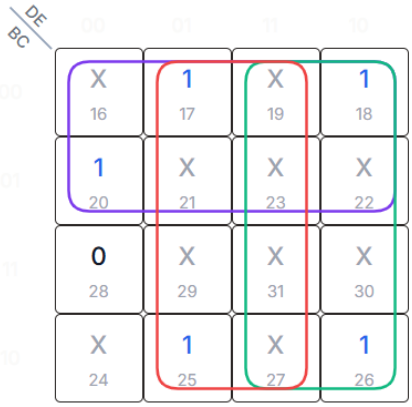
For implementing the finite state machine, the design requires a total of five state bits, denoted as P, Q, R, S, T. Since each bit corresponds to one state variable in the FSM, the system requires five flip-flops, one for each of the state bits (P, Q, R, S, T). These flip-flops store the present state and help transition the controller sequentially through all required Green and Yellow timing phases for each road direction.

K-Map Analysis for Next-State Logic:

For P/D1:



When A=0

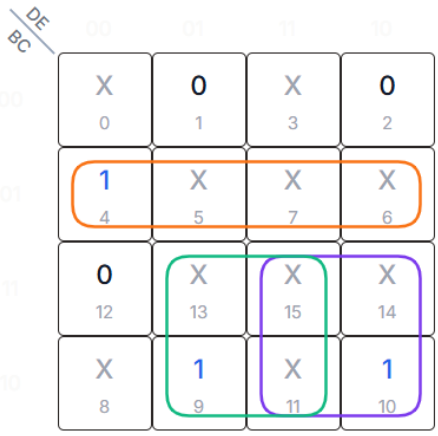


When A=1

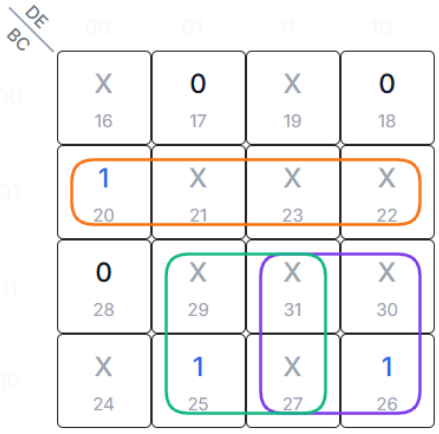
Output:

```
A'BC + AB' + AD + AE
```

For Q/D2:



When A=0



When A=1

Output:

```
B'C + BD + BE
```

For R/D3:

BC \ DE	00	01	11	10
00	X 0	0 1	X 3	1 2
01	0 4	X 5	X 7	X 6
11	0 12	X 13	X 15	X 14
10	X 8	0 9	X 11	1 10

When A=0

BC \ DE	00	01	11	10
00	X 16	0 17	X 19	1 18
01	0 20	X 21	X 23	X 22
11	0 28	X 29	X 31	X 30
10	X 24	0 25	X 27	1 26

When A=1

Output:

```
D
```

For S/D4:

BC \ DE	00	01	11	10
00	X 0	1 1	X 3	0 2
01	0 4	X 5	X 7	X 6
11	0 12	X 13	X 15	X 14
10	X 8	1 9	X 11	0 10

When A=0

BC \ DE	00	01	11	10
00	X 16	1 17	X 19	0 18
01	0 20	X 21	X 23	X 22
11	0 28	X 29	X 31	X 30
10	X 24	1 25	X 27	0 26

When A=1

Output:

```
E
```

For T/D5:

BC \ DE	00	01	11	10
00	X 0	0 1	X 3	0 2
01	1 4	X 5	X 7	X 6
11	1 12	X 13	X 15	X 14
10	X 8	0 9	X 11	0 10

When A=0

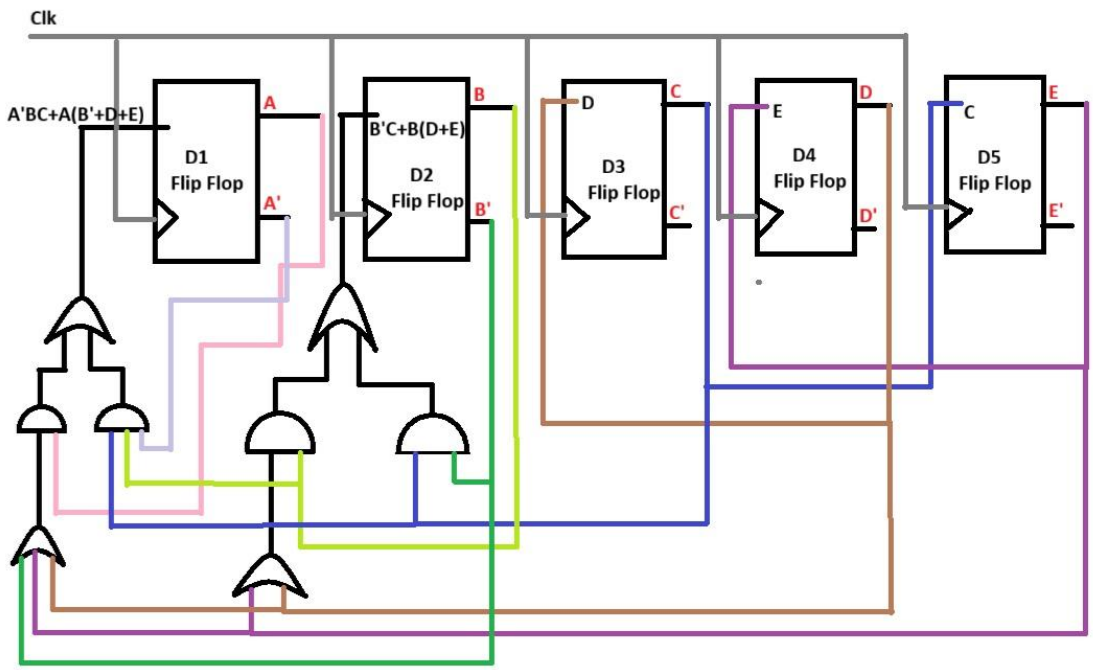
BC \ DE	00	01	11	10
00	X 16	0 17	X 19	0 18
01	1 20	X 21	X 23	X 22
11	1 28	X 29	X 31	X 30
10	X 24	0 25	X 27	0 26

When A=1

Output:

```
C
```

TRAFFIC LIGHT CONTROLLER CIRCUIT:



VERILOG CODE:

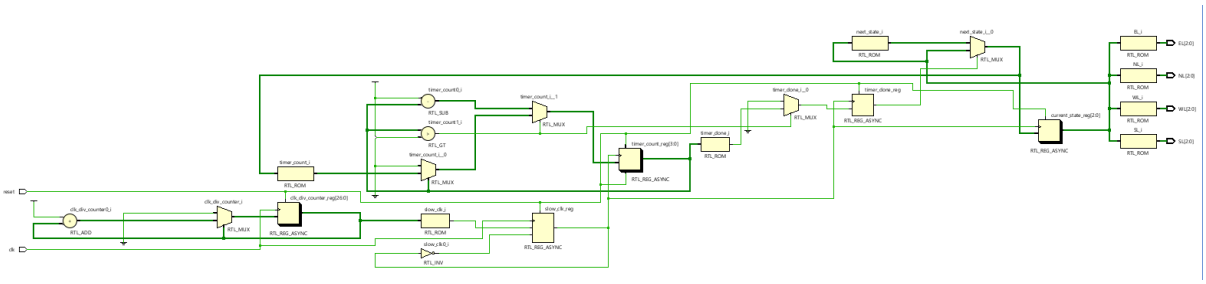
```
`timescale 1ns / 1ps

module traffic_light_controller (
    input wire clk,
    input wire reset,
    output reg [2:0] NL, // North Light
    output reg [2:0] SL, // South Light
    output reg [2:0] EL, // East Light
    output reg [2:0] WL  // West Light
);

    parameter N_G = 3'b000; // North Green
    parameter N_Y = 3'b001; // North Yellow
    parameter S_G = 3'b100; // South Green
    parameter S_Y = 3'b101; // South Yellow
    parameter E_G = 3'b010; // East Green
    parameter E_Y = 3'b011; // East Yellow
    parameter W_G = 3'b110; // West Green
```

```
parameter W_Y = 3'b111; // West Yellow

parameter GREEN = 3'b001;
parameter YELLOW = 3'b010;
parameter RED = 3'b100;

// Extending the clock to getvisible from nakes eye
localparam CLK_DIV_MAX = 27'd50000000;
reg [26:0] clk_div_counter = 27'd0;
reg slow_clk = 1'b0;

always @(posedge clk or posedge reset) begin
    if (reset) begin
        clk_div_counter <= 0;
        slow_clk <= 0;
    end else if (clk_div_counter == CLK_DIV_MAX - 1) begin
        clk_div_counter <= 0;
        slow_clk <= ~slow_clk;
    end else begin
        clk_div_counter <= clk_div_counter + 1;
    end
end

reg [2:0] current_state;
reg [2:0] next_state;

reg [3:0] timer_count;
reg timer_done;

localparam COUNT_GREEN = 4'd8;
localparam COUNT_YELLOW = 4'd4;

always @(posedge slow_clk or posedge reset) begin
    if (reset) begin
        current_state <= N_G; //Initially
    end else begin
        current_state <= next_state;
    end
end

// Counter
always @(posedge slow_clk or posedge reset) begin
    if (reset) begin
        timer_count <= COUNT_GREEN;
```

```
        timer_done <= 1'b0;
    end else if (timer_count > 1) begin
        timer_count <= timer_count - 1;
        timer_done <= 1'b0;
    end else if (timer_count == 1) begin
        timer_count <= timer_count - 1;
        timer_done <= 1'b1;
    end else begin
        timer_done <= 1'b0;

        case (next_state)
            N_G, S_G, E_G, W_G: timer_count <= COUNT_GREEN;
            N_Y, S_Y, E_Y, W_Y: timer_count <= COUNT_YELLOW;
            default: timer_count <= COUNT_GREEN;
        endcase
    end
end

always @(*) begin
    next_state = current_state;

    if (timer_done) begin
        case (current_state)
            N_G: next_state = N_Y;
            N_Y: next_state = S_G;

            S_G: next_state = S_Y;
            S_Y: next_state = E_G;

            E_G: next_state = E_Y;
            E_Y: next_state = W_G;

            W_G: next_state = W_Y;
            W_Y: next_state = N_G;
            default: next_state = N_G;
        endcase
    end
end

always @(*) begin
    NL = RED;
    SL = RED;
    EL = RED;
    WL = RED;
```

```
    case (current_state)
      N_G: NL = GREEN;
      N_Y: NL = YELLOW;

      S_G: SL = GREEN;
      S_Y: SL = YELLOW;

      E_G: EL = GREEN;
      E_Y: EL = YELLOW;

      W_G: WL = GREEN;
      W_Y: WL = YELLOW;

      default: begin
        NL = RED; SL = RED; EL = RED; WL = RED;
      end
    endcase
  end

endmodule\
```

SCHEMATIC:

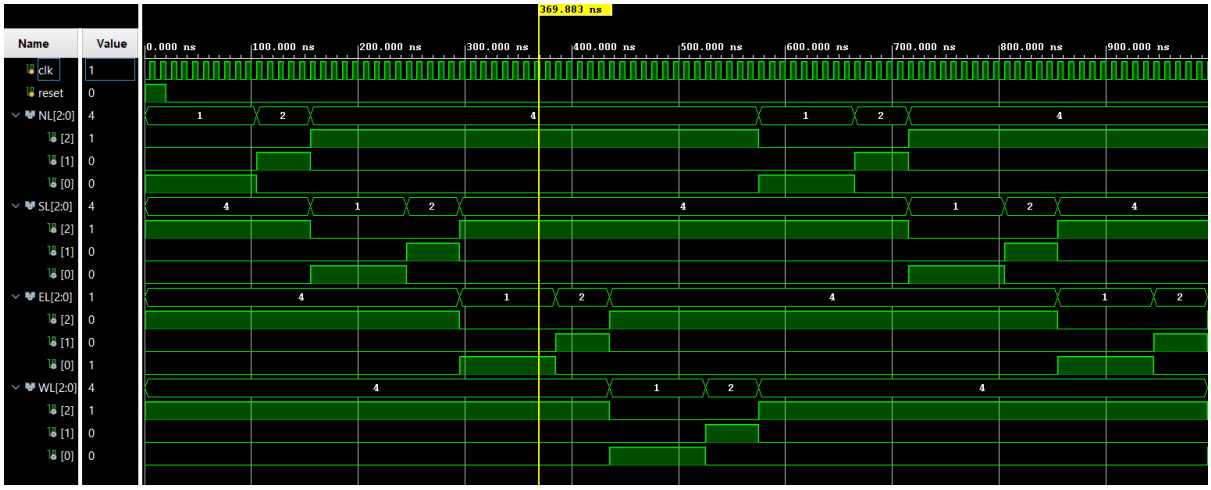


TESTBENCH:

```
`timescale 1ns / 1ps

module traffic_light_tb;

  reg clk;
  reg reset;
  wire [2:0] NL, SL, EL, WL;

  traffic_light_controller DUT (
    .clk(clk),
    .reset(reset),
    .NL(NL),
```

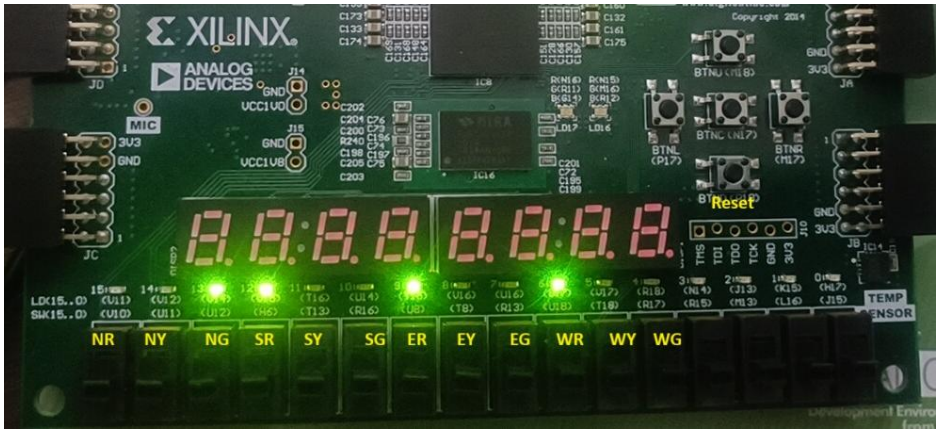
```
    .SL(SL),
    .EL(EL),
    .WL(WL)
  );

  initial begin
    clk = 0;
    forever #5 clk = ~clk;
  end

  initial begin
    reset = 1;
    #20;
    reset = 0;
  end

  initial begin
    $display("Time(ns)\t NL\t SL\t EL\t WL");
    $monitor("%0dns\t %b\t %b\t %b\t %b",
            $time, NL, SL, EL, WL);
  end

  initial begin
    #2000;
    $finish;
  end

endmodule
```

SIMULATION:

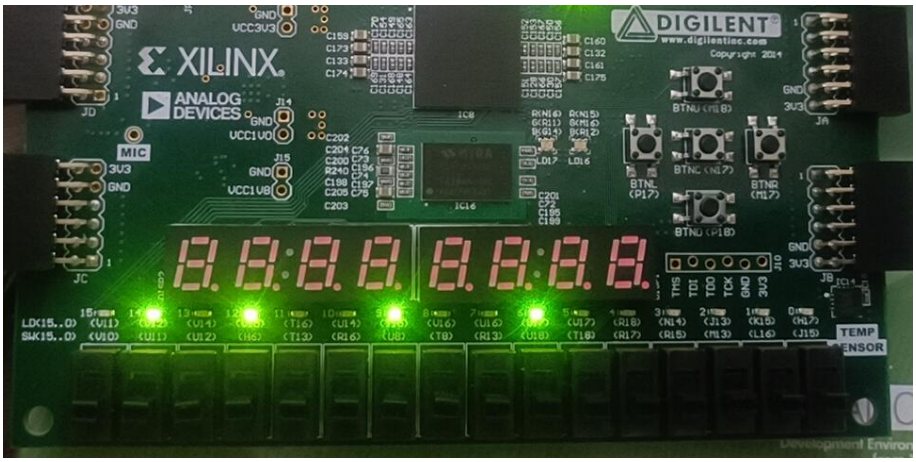


FPGA OUTPUT:

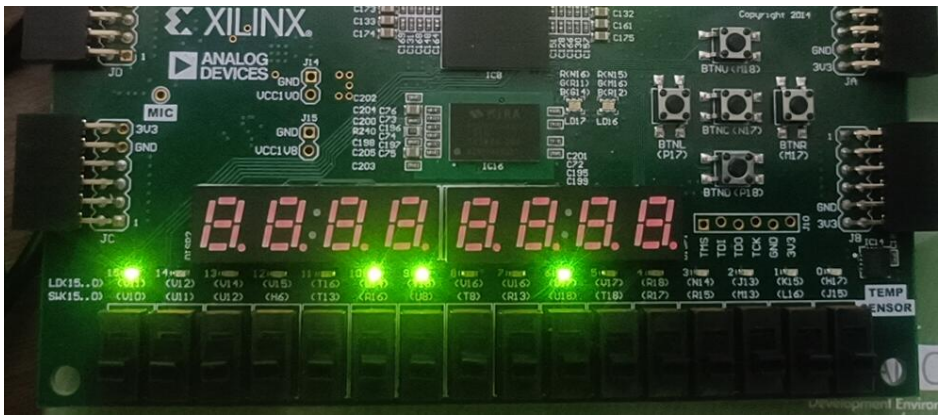
When North Light Green:



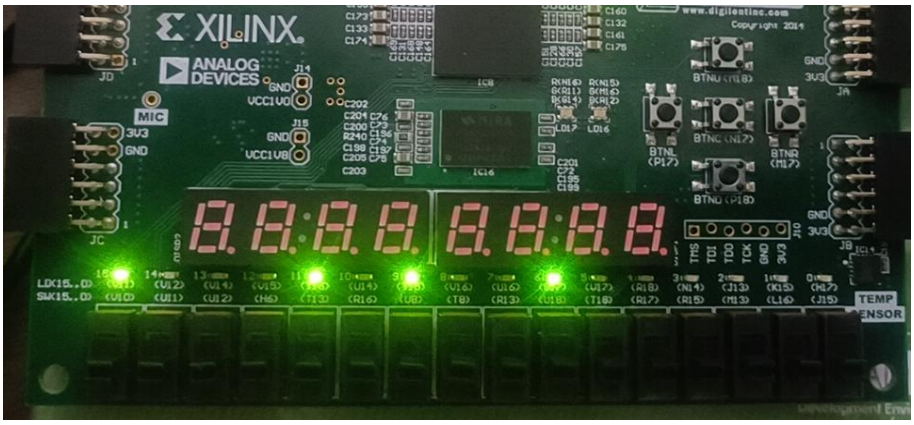
When North Light Yellow:



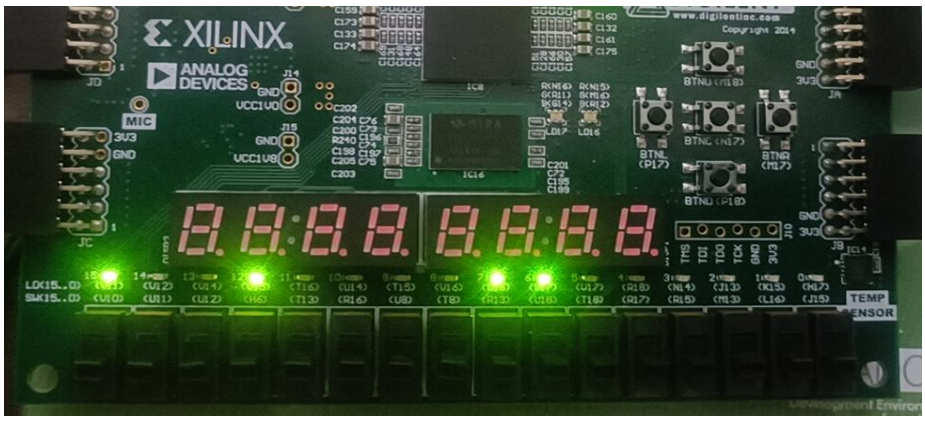
When South Light Green:



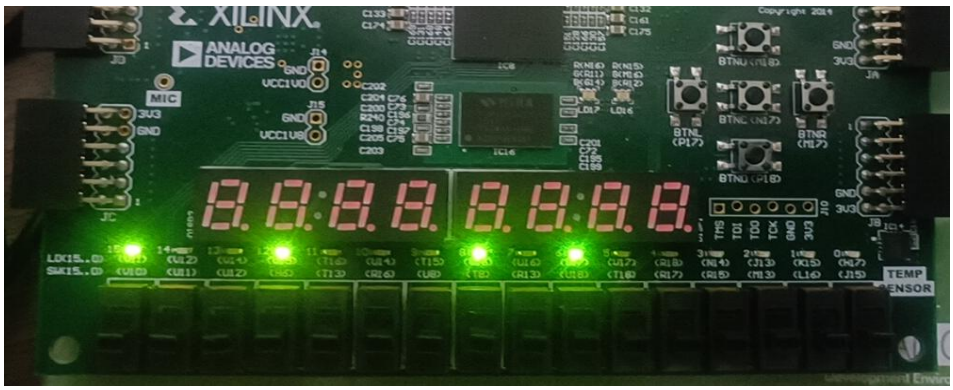
When South Light Yellow:



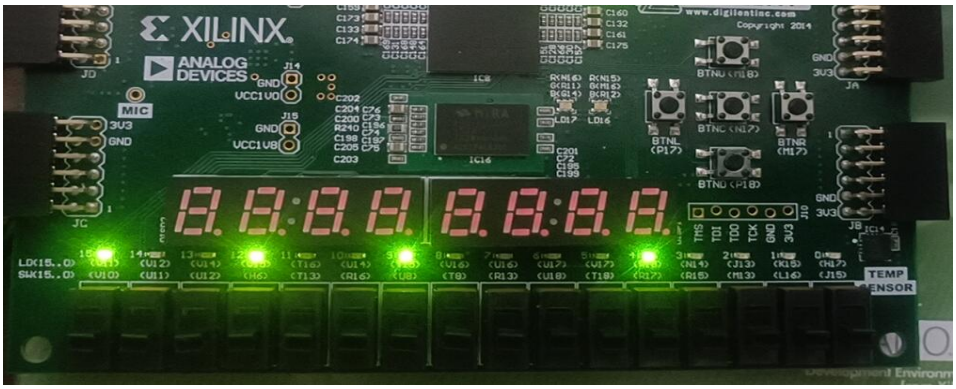
When East Light Green:



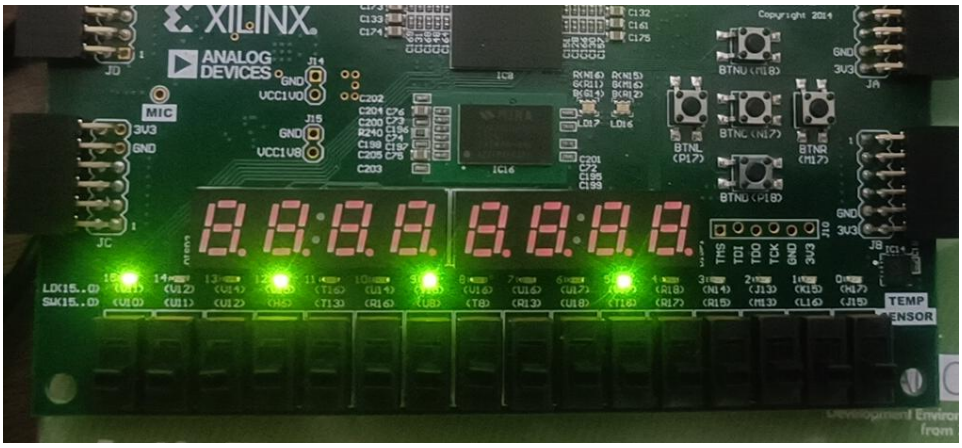
When East Light Yellow:



When West Light Green:



When West Light Yellow:



CONCLUSION:

The design and implementation of the Traffic Light Controller successfully demonstrate the practical application of digital system design concepts, particularly the use of finite state machines (FSMs), timing control, and sequential logic. By assigning a fixed duration to each traffic phase eight clock cycles of green followed by four clock cycles of yellow, the controller ensures smooth and systematic flow of traffic across all four directions. The encoding of signals, clear state transitions, and synchronous operation collectively contribute to a reliable and predictable design.

Through simulation, the functionality of the controller was verified, confirming correct timing behavior, accurate state progression, and proper light output for the North, South, East, and West directions. The project not only reinforces theoretical knowledge but also provides hands-on experience with hardware modeling, state machine implementation, and Verilog-based digital design. Overall, this work highlights how well-structured digital logic can be used to solve real-world problems such as traffic management while building a strong foundation for more complex control systems in the future.

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